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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Car Pricing Data Analysis

Project Domain

[Excel / Power BI]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project was developed as part of the **AICW (AI Career for Women) Fellowship Program**, a collaborative initiative by the **Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP**. The program aims to empower students by providing practical knowledge and hands-on experience in **Artificial Intelligence, Data Analytics, and Business Intelligence tools**.

The main objective of this project is to analyze and visualize automobile sales data to identify meaningful insights that can support better business decisions. The dataset used in this project contains information about various car attributes such as **brand, model, year, body type, seller details, state, and selling price**. Raw datasets often contain large volumes of information that are difficult to interpret without proper analysis and visualization.

To address this problem, the dataset was first examined and organized using **Microsoft Excel** to ensure that the data was structured and suitable for analysis. After preparing the data, **Microsoft Power BI** was used to build an interactive and visually appealing dashboard that presents important insights in a clear and understandable format.

Key performance indicators such as **average selling price, minimum selling price, maximum selling price, and total revenue** were calculated. Various visualizations including **pie charts, donut charts, and bar charts** were used to analyze car sales distribution by **state and body type**. Interactive filters were also added to allow users to explore the data based on **brand, year, and vehicle category**.

The final dashboard provides a comprehensive overview of the car sales data, helping users understand pricing patterns and regional sales performance. This project demonstrates how **data analytics and business intelligence tools** can transform raw data into valuable insights for decision-making.

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CHAPTER 1

Introduction

This project was undertaken as part of the **AI Career for Women (AICW) Fellowship Program**, organized by **Edunet Foundation** in collaboration with **Microsoft**, **LinkedIn**, and **SAP**. The fellowship program focuses on building practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence through real-world projects.

The project focuses on analyzing automobile sales data using modern data analytics tools. A raw dataset containing car-related information such as brand, model, year, state, body type, seller information, and selling price was obtained from Kaggle. Raw datasets often contain unstructured information, inconsistencies, and missing values, making them difficult to interpret directly.

To address this challenge, the dataset was first examined and cleaned using Microsoft Excel to ensure that the data was structured and suitable for analysis. After preprocessing, Microsoft Power BI was used to create an interactive dashboard that provides meaningful insights into car pricing trends, regional sales distribution, and vehicle category performance. The dashboard includes key metrics such as average selling price, minimum and maximum selling price, and total revenue, along with visualizations that allow users to explore the data interactively.

Through this project, the aim is to demonstrate how data analytics and business intelligence tools can convert raw datasets into meaningful insights that support data-driven decision-making.

1.1 Problem Statement

In the automobile industry, large volumes of sales data are generated from different sellers, locations, and vehicle categories. However, this data is often stored in raw and unstructured formats, making it difficult for businesses to analyze pricing trends and sales performance effectively.

The car price dataset used in this project contains multiple attributes such as vehicle brand, model year, state, body type, seller information, and selling price. Without proper data cleaning and visualization, it becomes challenging to identify patterns such as which states generate higher sales, which body types are most popular, and how vehicle prices vary across different categories.

This problem is significant because automobile dealers, analysts, and business managers rely on accurate data insights to make informed decisions regarding pricing strategies, inventory management, and market demand. Poorly structured data can lead to incorrect analysis and ineffective business strategies.

Existing approaches often rely on manual analysis using spreadsheets, which can be time-consuming and prone to errors when dealing with large datasets. Therefore, there is a need for automated and interactive analytical tools that can transform raw data into clear and meaningful insights.

This project addresses this challenge by cleaning the dataset and building an interactive dashboard that allows users to easily explore car price trends and sales distribution across different categories.

1.2 Motivation

The motivation behind this project comes from the increasing importance of data analytics in modern businesses. Organizations today rely heavily on data to understand market trends, customer preferences, and product performance.

Analyzing automobile sales data provides valuable insights that can help businesses improve pricing strategies and identify high-performing markets. By applying data analytics techniques to the car price dataset, it becomes possible to identify patterns such as which states have higher sales activity and which vehicle body types generate higher revenue.

This project is also personally meaningful as it provides practical experience in using industry-relevant tools such as Microsoft Excel and Power BI for data analysis and visualization. Developing interactive dashboards helps in understanding how real-world data can be transformed into actionable insights.

Furthermore, this project aligns with current industry trends where businesses are increasingly adopting artificial intelligence and business intelligence tools to support data-driven decision-making.

1.3 Objectives

The key objectives of this project are:

1. To collect the automobile sales dataset from Kaggle and understand its structure and variables.

2. To clean and preprocess the raw data using Microsoft Excel by removing duplicates, correcting formats, and handling missing values.

3. To perform data analysis to identify patterns in car selling prices across different states, sellers, and body types.

4. To develop an interactive dashboard using Microsoft Power BI to visualize key insights and trends.

5. To calculate important business metrics such as average selling price, minimum price, maximum price, and total revenue.

6. To present the findings through clear and interactive visualizations that support better understanding and decision-making.

1.4 Scope of the Project

This project focuses on analyzing and visualizing automobile sales data to extract meaningful insights about car pricing and sales distribution. The dataset used in this project consists of **car price records obtained from Kaggle**, which includes information such as vehicle brand, model year, body type, seller details, state, and selling price.

The project workflow involves collecting the raw dataset, performing data cleaning and preprocessing using **Microsoft Excel**, and building an interactive analytical dashboard using **Microsoft Power BI**. Excel is used for preparing and organizing the dataset, while Power BI is used to create visualizations and perform data analysis.

The analysis in this project mainly focuses on **data cleaning, exploratory data analysis, and dashboard visualization**. Various key performance indicators such as **average selling price, minimum selling price, maximum selling price, and total revenue** are calculated to understand the overall performance of car sales. The dashboard also provides insights into sales distribution based on **state, body type, brand, and year**.

However, the scope of this project is limited to **descriptive data analysis and visualization**. The project does not include advanced features such as **real-time data integration, predictive machine learning models, or automated data pipelines**.

This project is developed as part of the **AI Career for Women (AICW) Fellowship Program** and is intended to demonstrate practical skills in **data analytics and business intelligence tools**. Therefore, it serves as a **learning and showcase project rather than a full production-level business system**.

CHAPTER 2

Literature Survey

2.1 Existing Work and Research

Several studies and industry practices have focused on analyzing business and sales data using business intelligence tools. Many organizations use dashboards and reporting tools to analyze large datasets and extract meaningful insights for decision-making. Research and documentation related to data analytics emphasize the importance of data cleaning, transformation, and visualization before performing analysis.

Automobile sales data analysis has been widely used in the business domain to understand market trends, customer preferences, and pricing patterns. Analysts often use structured datasets containing information such as vehicle brand, model year, body type, location, and selling price to identify trends and evaluate sales performance.

Previous approaches often relied on spreadsheet tools for basic data analysis and reporting. While spreadsheet tools allow users to organize and filter data, they have limitations when handling large datasets or creating advanced interactive visualizations. Modern business intelligence tools provide more advanced capabilities such as dynamic dashboards, automated calculations, and interactive filtering.

This project uses a car price dataset obtained from **Kaggle**, which contains detailed records of automobile sales. The dataset includes attributes such as brand, model year, body type, seller, state, and selling price. By applying modern business intelligence techniques, this project aims to transform raw data into clear visual insights through an interactive dashboard.

2.2 Tools and Technologies Review

Several tools and technologies were used to complete this project. These tools help in organizing the dataset, performing analysis, and creating interactive dashboards.

Microsoft Excel
Microsoft Excel is widely used for data cleaning, data transformation, and preliminary analysis. It provides features such as sorting, filtering, removing duplicates, and formatting data. Excel helps prepare the dataset so that it can be used effectively in further analytical tools.

Microsoft Power BI
Microsoft Power BI is a powerful business intelligence tool used for creating interactive dashboards and reports. It allows users to connect datasets, create data models, build visualizations, and calculate key performance indicators. Power BI also enables users to explore data through filters, slicers, and dynamic charts.

Microsoft Copilot
Microsoft Copilot is an AI-powered assistant integrated into Microsoft tools that helps automate tasks, generate insights, and assist users in analyzing data using natural language prompts.

ChatGPT and **Google Gemini**
AI chatbots such as ChatGPT and Google Gemini are generative AI tools that assist in content creation, report writing, and explanation of analytical results. These tools help users structure reports and interpret analytical findings more effectively.

Together, these tools enable efficient data processing, analysis, and presentation of insights.

2.3 Gaps Identified

Although many studies and solutions exist for data analytics and business intelligence, certain gaps remain in practical implementation, especially for beginners and students.

Most existing research focuses on advanced machine learning techniques for predictive analysis, which may require complex programming knowledge. However, simpler analytical solutions using tools such as Excel and Power BI are often more practical for students and entry-level analysts.

Additionally, there is limited documentation on how AI assistants like Microsoft Copilot and AI chatbots can be effectively integrated with traditional tools such as Excel and Power BI in student-level projects.

Another gap is that many datasets available online, such as those on Kaggle, are raw and unstructured. Without proper data cleaning and visualization, it becomes difficult to extract meaningful insights from these datasets.

This project addresses these gaps by demonstrating how raw automobile sales data can be cleaned, analyzed, and visualized using accessible business intelligence tools. The project highlights a practical workflow that transforms raw data into an interactive dashboard that supports data-driven understanding and decision-making.

CHAPTER 3

Proposed Methodology

3.1 System Design

This project follows a structured data-analysis workflow to convert raw car-price records into meaningful business insights using Power BI.

Data in this project comes from a large automobile sales dataset containing information such as year, make, model, trim, mileage, condition, state, color, MMR value, and selling price. The system processes this raw data and generates an interactive Power BI dashboard.

Data Flow:
• Data Source / Input:
The input dataset is an Excel file containing used-car transactional records. Each row represents one vehicle with attributes:
YEAR, MAKE, MODEL, TRIM, BODY, TRANSMISSION, VIN, STATE, CONDITION, ODOMETER, COLOR, INTERIOR, SELLER, MMR, SELLINGPRICE, and SALEDATE.
The dataset includes hundreds of vehicle entries from brands like Kia, BMW, Chevrolet, Audi, Nissan, Hyundai, Toyota, Mercedes-Benz, etc.

• Data Cleaning:
The following steps were performed:
– Removed blank VIN values and unnecessary special characters
– Fixed incorrect spellings and inconsistent states (ex: “skikkim” → “sikkim”)
– Standardized columns such as BODY, COLOR and TRANSMISSION
– Converted SALEDATE to proper Date format
– Removed extreme outliers where ODOMETER values were unrealistic (e.g., 999999)
– Ensured numeric fields (MMR, SELLINGPRICE, ODOMETER) used correct data types
– Removed duplicate rows based on VIN
– Ensured missing TRIM or MODEL values were filled where possible

• Analysis / Processing:
Power BI and Excel tools were used to analyse the dataset:
– Created calculated fields such as:

- Profit = SELLINGPRICE − MMR
- Average Selling Price per Make
- Total Sales by State
- Car Condition Score Analysis
  – Generated DAX measures including:
- Total Sales
- Average Price

- Average Mileage

- Count of Vehicles Sold
  – Applied filters and slicers by Year, Make, Body Type, and State
  – Used Power BI visuals for trend and comparison analysis
  – Used AI prompts (Copilot) to generate insights and summaries

• Visualization / Output:
The final result is an interactive Power BI dashboard containing:
– Total Cars Sold
– Average Selling Price
– Sales by Car Make
– Top-Selling States
– Price vs Odometer Scatter Plot
– Condition vs Selling Price Relationship
– Profitability Analysis (Selling Price vs MMR)
– Car Body Type Distribution
– Monthly/Yearly Sales Trend

Charts, KPIs, and slicers allow easy exploration of car-market performance.

3.2 Modules / Components Used

The project consists of the following modules:

• Module 1 — Data Ingestion
Importing raw car-price Excel data into Power BI Desktop.

• Module 2 — Data Cleaning
Removing inconsistent VINs, correcting states, normalizing car models, converting data types, removing outliers, and standardizing Date fields.

• Module 3 — Data Analysis
Using Power BI DAX measures to calculate averages, totals, profit, and trends. Preparing detailed tables for Make-wise and State-wise comparison.

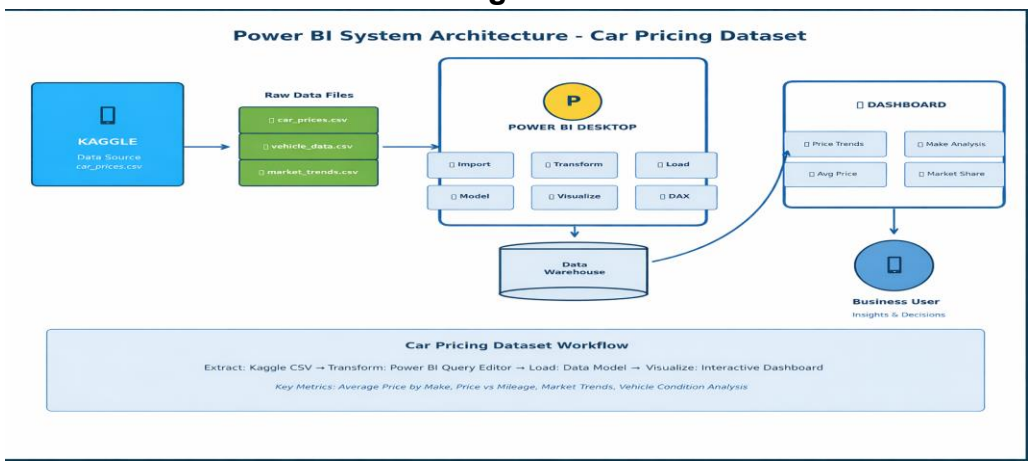
• Module 4 — Visualization / Report Generation
Building an interactive dashboard with charts including bar charts, KPI cards, donut charts, line graphs, matrix tables, and maps.

• Module 5 — AI Integration
Using Microsoft Copilot and ChatGPT to summarize trends, generate explanations, detect anomalies, and convert insights into readable reports.

• Module 6 — Insight Generation & Reporting
Extracting final findings such as highest-selling brand, price trends, states with maximum demand, and profit comparisons.

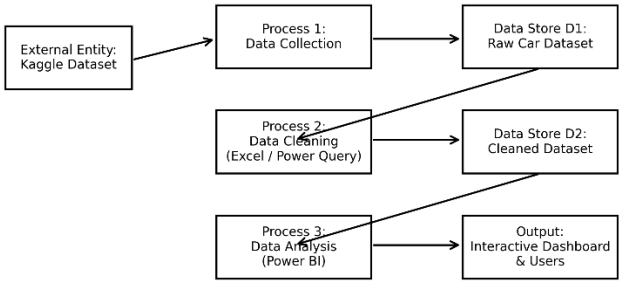
3.3 Data Flow Diagram (DFD)

DFD Level 0 — Context Diagram



Raw Car Data → (Car Price Analysis System) → Cleaned Dataset / Power BI Dashboard Output

DFD Level 1 — Detailed Process Flow
- Sub-process 1: Data Loading & Validation
- Sub-process 2: Data Cleaning & Transformation
- Sub-process 3: Data Analysis (Excel / DAX / AI)
- Sub-process 4: Dashboard Creation & Report Publishing



Level-1 Data Flow Diagram (DFD) for Car Price Analysis Dashboard

3.4 Advantages

List the key advantages of your proposed approach:

- Automates car-price analysis by eliminating manual calculation work
- Enables quick insights on selling patterns, brand performance, and pricing trends
- Uses AI to reduce human error and speed up exploratory analysis
- Scalable for large datasets with thousands of vehicle entries
- Cost-effective since tools like Excel and Power BI are included in Microsoft 365
- Helps dealers understand market trends and optimize pricing strategies
- Visual dashboard improves decision making for non-technical stakeholders

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	AMD Ryzen 5 3550H with Radeon Vega Mobile Gfx
RAM	16.0 GB (13.9 GB usable)
Storage	466 GB SSD Samsung SSD 980 500GB
Display	1920*1080 resolution or higher
Internet	Required for Copilot , power BI services and AI tools

Software Requirements

Software	Details
Operating System	Windows 10
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results.

4.1 Data Cleaning Results

The first step in the project involved cleaning and preparing the dataset so that it could be used effectively for analysis and visualization.

Dataset Source:
The dataset used in this project was downloaded from **Kaggle**, which provides publicly available datasets for data analysis and machine learning projects. The dataset contains automobile sales information including car brand, model year, body type, state, seller details, and selling price.

Original Dataset Size:
The raw dataset initially contained multiple records with several columns describing different car attributes such as brand, year, body type, seller, state, and price.

Issues Found:
During the initial exploration of the dataset, several issues were identified:

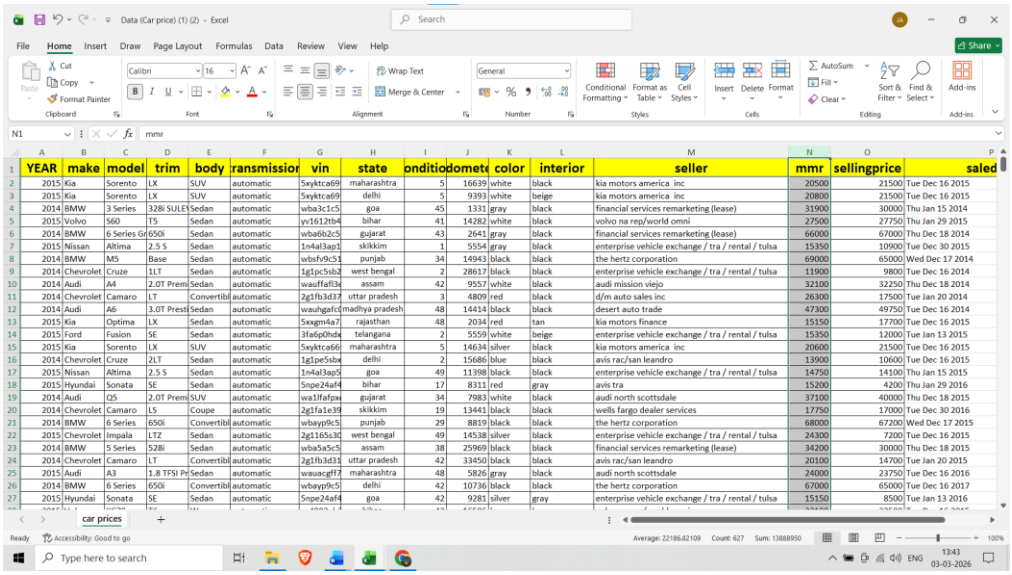
- Duplicate records present in the dataset
- Missing or null values in some columns
- Inconsistent formatting in certain fields
- Data types that needed correction for proper analysis

Cleaning Steps Applied:
The following steps were performed to clean the dataset using **Microsoft Excel**:

1. Removed duplicate records using the **Remove Duplicates** feature.
2. Checked for missing or blank values and corrected them where necessary.
3. Standardized formatting across columns to ensure consistency.
4. Verified data types for numerical and categorical fields.
5. Organized the dataset into a structured format suitable for importing into **Microsoft Power BI**.

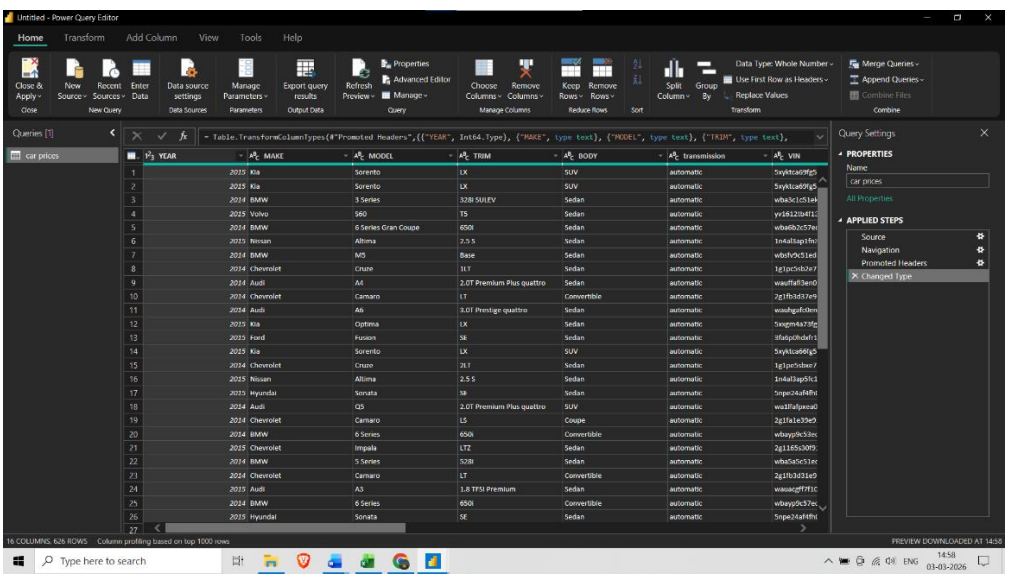
After completing these steps, the dataset became structured and ready for further analysis and visualization.

Cleaned Dataset Size:
After the cleaning process, the dataset contained consistent and validated records suitable for building the analytical dashboard.



YEAR	make	model	trim	body	ransmissior	vin	state	onditio	domete	color	interior	seller	mmr	sellingprice	saled
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69	maharashtra	5	16639	white	black	kia motors america inc	20500	21500	Tue Dec 16 2015
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69	delhi	5	9393	white	beige	kia motors america inc	20800	21500	Tue Dec 16 2015
2014	BMW	3 Series	328i SULE	Sedan	automatic	wba3c1c5	goa	45	1331	gray	black	financial services remarketing (lease)	31900	30000	Thu Jan 15 2014
2015	Volvo	S60	T5	Sedan	automatic	yv1612tb4	bihar	41	14282	white	black	volvo na rep/world omni	27500	27750	Thu Jan 29 2015
2014	BMW	6 Series Gr	650i	Sedan	automatic	wba6b2c5	gujarat	43	2641	gray	black	financial services remarketing (lease)	66000	67000	Thu Dec 18 2014
2015	Nissan	Altima	2.5 S	Sedan	automatic	1n4al3ap1	skikkim	1	5554	gray	black	enterprise vehicle exchange / tra / rental / tulsa	15350	10900	Tue Dec 30 2015
2014	BMW	M5	Base	Sedan	automatic	wbsfv9c5	punjab	34	14943	black	black	the hertz corporation	69000	65000	Wed Dec 17 2014
2014	Chevrolet	Cruze	1LT	Sedan	automatic	1g1pc5sb2	west bengal	2	28617	black	black	enterprise vehicle exchange / tra / rental / tulsa	11900	9800	Tue Dec 16 2014
2014	Audi	A4	2.0T Premi	Sedan	automatic	wauffaf3e	assam	42	9557	white	black	audi mission viejo	32100	32250	Thu Dec 18 2014
2014	Chevrolet	Camaro	LT	Convertibl	automatic	2g1fb3d37	uttar pradesh	3	4809	red	black	d/m auto sales inc	26300	17500	Tue Jan 20 2014
2014	Audi	A6	3.0T Presti	Sedan	automatic	wauhgafc	madhya pradesh	48	14414	black	black	desert auto trade	47300	49750	Tue Dec 16 2014
2015	Kia	Optima	LX	Sedan	automatic	5xxgm4a7	rajasthan	48	2034	red	tan	kia motors finance	15150	17700	Tue Dec 16 2015
2015	Ford	Fusion	SE	Sedan	automatic	3fa6p0hd	telangana	2	5559	white	beige	enterprise vehicle exchange / tra / rental / tulsa	15350	12000	Tue Jan 13 2015
2015	Kia	Sorento	LX	SUV	automatic	5xyktca66	maharashtra	5	14634	silver	black	kia motors america inc	20600	21500	Tue Dec 16 2015
2014	Chevrolet	Cruze	2LT	Sedan	automatic	1g1pe5sbx	delhi	2	15686	blue	black	avis rac/san leandro	13900	10600	Tue Dec 16 2015
2015	Nissan	Altima	2.5 S	Sedan	automatic	1n4al3ap5	goa	49	11398	black	black	enterprise vehicle exchange / tra / rental / tulsa	14750	14100	Thu Jan 15 2015
2015	Hyundai	Sonata	SE	Sedan	automatic	5npe24af4	bihar	17	8311	red	gray	avis tra	15200	4200	Thu Jan 29 2016
2014	Audi	Q5	2.0T Premi	SUV	automatic	wa1lfafpa	gujarat	34	7983	white	black	audi north scottsdale	37100	40000	Thu Dec 18 2015
2014	Chevrolet	Camaro	LS	Coupe	automatic	2g1fa1e39	skikkim	19	13441	black	black	wells fargo dealer services	17750	17000	Tue Dec 30 2016
2014	BMW	6 Series	650i	Convertibl	automatic	wbayp9c5	punjab	29	8819	black	black	the hertz corporation	68000	67200	Wed Dec 17 2015
2015	Chevrolet	Impala	LTZ	Sedan	automatic	2g1165s3d	west bengal	49	14538	silver	black	enterprise vehicle exchange / tra / rental / tulsa	24300	7200	Tue Dec 16 2015
2014	BMW	5 Series	528i	Sedan	automatic	wba5a5c5	assam	38	25969	black	black	financial services remarketing (lease)	34200	30000	Thu Dec 18 2015
2014	Chevrolet	Camaro	LT	Convertibl	automatic	2g1fb3d31	uttar pradesh	42	33450	black	black	avis rac/san leandro	20100	14700	Tue Jan 20 2015
2015	Audi	A3	1.8 TFSI Pr	Sedan	automatic	wauacgff7	maharashtra	48	5826	gray	black	audi north scottsdale	24000	23750	Tue Dec 16 2016
2014	BMW	6 Series	650i	Convertibl	automatic	wbayp9c5	delhi	42	10736	black	black	the hertz corporation	67000	65000	Tue Dec 16 2017
2015	Hyundai	Sonata	SE	Sedan	automatic	5npe24af4	goa	42	9281	silver	gray	enterprise vehicle exchange / tra / rental / tulsa	15150	8500	Tue Jan 13 2016

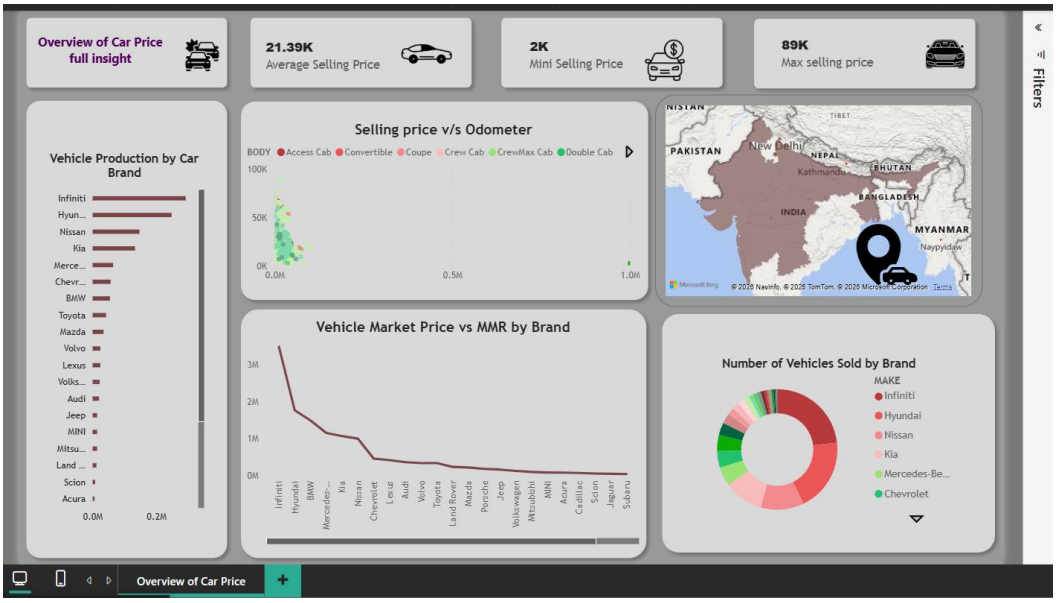
Excel Dataset (uncleaned dataset)



Power Query Data Transformation Image

4.2 Visualization / Output Results

- After cleaning and preparing the dataset, an interactive dashboard was developed using **Microsoft Power BI** to visualize the key insights from the car price dataset. The dashboard provides a comprehensive overview of vehicle pricing, brand distribution, and market trends.
- **Chart 1: Key Performance Indicators (KPI Cards)**
  At the top of the dashboard, several KPI cards were created to summarize the most important metrics from the dataset. These include the **average selling price**, **minimum selling price**, and **maximum selling price** of vehicles. These indicators provide a quick overview of the overall price distribution within the dataset and allow users to understand the general pricing trends in the automobile market.
- **Chart 2: Vehicle Production by Car Brand**
  A horizontal bar chart is used to display the number of vehicles available for each car brand. This visualization helps identify which car manufacturers appear most frequently in the dataset and provides insights into brand presence in the market.
- **Chart 3: Selling Price vs Odometer**
  A scatter plot visualization is used to analyze the relationship between **vehicle odometer readings** and **selling prices**. This chart helps understand how vehicle mileage affects the selling price of cars. Typically, vehicles with lower mileage tend to have higher selling prices, while higher mileage vehicles generally have lower market value.
- **Dashboard / Chart 4: Geographic Distribution of Vehicles by State**
  A map visualization was created to show the distribution of vehicle sales across different states. This visualization allows users to easily identify regional patterns and understand which locations contribute more to the dataset.
- **Chart 5: Vehicle Market Price vs MMR by Brand**
  A line chart was used to compare the **market selling price** with **Manheim Market Report (MMR) values** across different vehicle brands. This visualization helps analyze price differences and evaluate whether vehicles are sold above or below their estimated market value.
- **Chart 6: Number of Vehicles Sold by Brand**
  A donut chart was created to represent the distribution of vehicles sold by brand. Each segment of the chart represents a specific manufacturer, allowing users to quickly compare brand presence within the dataset.
- These visualizations collectively provide a detailed overview of the automobile dataset, helping users understand pricing patterns, brand distribution, and regional vehicle sales trends.



Car Price Analysis Interactive Dashboard in Microsoft Power BI

4.3 AI Tool Integration Results

Artificial Intelligence tools were also used during the project to assist with analysis explanations and report preparation.

Tool Used:
The AI tool used in this project was **ChatGPT**, which helped generate explanations, summarize insights, and assist in structuring the project report.

Prompts Used:
Some examples of prompts used during the project include:

- "Explain the Power BI dashboard created for car price analysis."
- "Generate a project abstract for a car price dataset analysis project."
- "Help write a literature survey and methodology section for a data analytics project."

Output Quality:
The AI-generated responses were clear, structured, and helpful for preparing the documentation of the project. The tool helped simplify complex explanations and organize the content in a professional format.

How It Enhanced the Project:
Using AI tools improved the efficiency of report writing and helped in generating well-structured explanations of the analysis. It reduced the time required to prepare documentation and supported better presentation of the project results.

Microsoft 365
Copilot

Gemini

PowerBI

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

The project focused on analyzing a car price dataset and creating an interactive dashboard using Power BI. After cleaning and transforming the dataset, several meaningful insights were discovered through data visualization and analysis.

• After performing data cleaning in **Microsoft Power BI**, duplicate and inconsistent records were removed, which improved the accuracy and reliability of the dataset used for analysis.

• The analysis revealed that vehicles with **lower odometer readings generally have higher selling prices**, indicating that mileage is one of the most important factors affecting the resale value of cars.

• The dashboard showed that **some car brands appear more frequently in the dataset**, suggesting that these brands have a higher presence in the used car market.

• Through the comparison between **selling price and Manheim Market Report (MMR)** values, it was observed that some vehicles are sold above the estimated market value while others are sold below it.

• Using AI tools such as **ChatGPT**, **Microsoft Copilot**, and **Google Gemini** helped speed up documentation, data understanding, and report preparation during the project.

• The interactive dashboard provides a **clear visual representation of car prices, mileage distribution, brand popularity, and geographic vehicle distribution**, making it easier for users to understand the dataset.

5.2 Project Links

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project: [Paste your GitHub repository link here]
- Video Recording of Project Demonstration: [google drive]

5.3 Limitations

Although the project successfully achieved its objectives, there were some limitations.

• The project used a **sample dataset**, which may not represent all real-world used car market conditions.

• The dashboard is based on **static data**, meaning it does not automatically update with real-time vehicle market data.

• Some AI-assisted features of **Microsoft Copilot** were limited during the project due to restricted access.

• The analysis focuses mainly on **price, brand, mileage, and location**, while other important factors such as vehicle condition or accident history were not included in the dataset.

5.4 Future Work

Several improvements can be made to enhance the project in the future.

• Integrate **real-time data sources or APIs** to automatically update the dataset in Power BI.

• Apply **machine learning models** to predict future car prices based on historical data.

• Expand the dataset to include additional attributes such as **fuel type, vehicle condition, and accident history**.

• Develop a **web-based interface or mobile application** that allows users to explore car price insights easily.

• Improve the dashboard by adding more advanced visualizations and interactive filters.

• Create a **standardized AI prompt system** to improve the efficiency of AI-assisted data analysis.

5.5 Conclusion

This project successfully demonstrated how data analysis and visualization tools can be used to understand trends in the used car market. Using **Microsoft Power BI**, the dataset was cleaned, transformed, and visualized to create an interactive dashboard that highlights key insights such as vehicle price distribution, mileage impact, and brand presence in the market.

The use of AI tools such as **ChatGPT**, **Microsoft Copilot**, and **Google Gemini** helped support research, report writing, and understanding of technical concepts throughout the project. These tools significantly improved productivity and simplified the process of documenting the analysis.

All project objectives were achieved, including data preparation, analysis, visualization, and dashboard development. The final dashboard provides meaningful insights that can help users better understand vehicle pricing trends and market patterns.

Overall, this project enhanced practical knowledge in **data analytics, business intelligence, and AI-assisted workflows**, making it a valuable learning experience for applying analytical skills to real-world datasets.

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APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Sample Car Price Dataset in Microsoft Excel

YEAR	MAKE	MODEL	TRIM	BODY	transmissi	VIN	STATE	CONDITION	ODOMETER	COLOR	INTER
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg566472	maharashtra	5	16639	white	black
2015	Kia	Sorento	LX	SUV	automatic	5xyktca69fg561319	delhi	5	9393	white	beige
2014	BMW	3 Series	328i SULEV	Sedan	automatic	wba3c1c51ek116351	goa	45	1331	gray	black
2015	Volvo	S60	T5	Sedan	automatic	yv1612tb4f1310987	bihar	41	14282	white	black
2014	BMW	6 Series Gran (	650i	Sedan	automatic	wba6b2c57ed129731	gujarat	43	2641	gray	black
2015	Nissan	Altima	2.5 S	Sedan	automatic	1n4al3ap1fn326013	skikkim	1	5554	gray	black
2014	BMW	M5	Base	Sedan	automatic	wbsfv9c51ed593089	punjab	34	14943	black	black
2014	Chevrolet	Cruze	1LT	Sedan	automatic	1g1pc5sb2e7128460	west bengal	2	28617	black	black
2014	Audi	A4	2.0T Premi	Sedan	automatic	wauffafl3en030343	assam	42	9557	white	black
2014	Chevrolet	Camaro	LT	Convertibl	automatic	2g1fb3d37e9218789	uttar pradesh	3	4809	red	black
2014	Audi	A6	3.0T Presti	Sedan	automatic	wauhgafc0en062916	madhya pradesh	48	14414	black	black
2015	Kia	Optima	LX	Sedan	automatic	5xxgm4a73fg353538	rajasthan	48	2034	red	tan
2015	Ford	Fusion	SE	Sedan	automatic	3fa6p0hdxfr145753	telangana	2	5559	white	beige
2015	Kia	Sorento	LX	SUV	automatic	5xyktca66fg561407	maharashtra	5	14634	silver	black
2014	Chevrolet	Cruze	2LT	Sedan	automatic	1g1pe5sbxe7120097	delhi	2	15686	blue	black
2015	Nissan	Altima	2.5 S	Sedan	automatic	1n4al3ap5fc124223	goa	49	11398	black	black
2015	Hyundai	Sonata	SE	Sedan	automatic	5npe24af4fh001562	bihar	17	8311	red	gray
2014	Audi	Q5	2.0T Premi	SUV	automatic	wa1lfafpxea085074	gujarat	34	7983	white	black
2014	Chevrolet	Camaro	LS	Coupe	automatic	2g1fa1e39e9134494	skikkim	19	13441	black	black
2014	BMW	6 Series	650i	Convertibl	automatic	wbayp9c53ed169260	punjab	29	8819	black	black
2015	Chevrolet	Impala	LTZ	Sedan	automatic	2g1165s30f9103921	west bengal	49	14538	silver	black
2014	BMW	5 Series	528i	Sedan	automatic	wba5a5c51ed501631	assam	38	25969	black	black
2014	Chevrolet	Camaro	LT	Convertibl	automatic	2g1fb3d31e9134662	uttar pradesh	42	33450	black	black

Below is a sample of the raw dataset used in this project. The dataset contains information about different vehicles including their manufacturing year, brand, model, transmission type, location, mileage, seller details, market price (MMR), and selling price.

The dataset was obtained from **Kaggle** and later cleaned and prepared before importing it into **Microsoft Power BI** for analysis and dashboard visualization.

Appendix B — Key Excel Formulas / DAX Measures Used

Power BI DAX Measures

• **Average Sale Price = AVERAGE(CarData[SellingPrice])**
Used to calculate the **average selling price of all vehicles** displayed in the KPI card.

• **Total Vehicles = COUNT(CarData[VehicleID])**
Used to calculate the **total number of vehicles in the dataset**.

• **Max Price = MAX(CarData[SellingPrice])**
Used to identify the **highest priced vehicle**.

• **Min Price = MIN(CarData[SellingPrice])**
Used to identify the **lowest priced vehicle**.

• **Total Brand = DISTINCTCOUNT(CarData[Make])**
Used to calculate the **total number of unique car brands**, displayed in the dashboard KPI.

Appendix C — AI Prompts Used

During the development of the **Car Pricing Analysis Dashboard**, AI tools such as Copilot or an AI Chatbot were used to assist with data analysis, formula selection, and dashboard design. The following prompts were used to guide the AI in completing different tasks during the project.

• **Prompt 1:**
"Analyze this car pricing dataset and summarize the key insights such as average selling price, total number of brands, and number of car models."

• **Prompt 2:**
"Suggest suitable charts and visualizations for a car price analysis dashboard using fields like brand, selling price, odometer, body type, and market price."

• **Prompt 3:**
"Create Excel formulas to calculate the average, minimum, and maximum selling price from the car pricing dataset."

• **Prompt 4:**
"Help design a Power BI dashboard layout to show vehicle production by brand, selling price vs odometer, and number of vehicles sold by brand."

• **Prompt 5:**
"Explain how to create KPI cards in Power BI to display total brands, average sale price, and number of car models."

• **Prompt 6:**
"Identify possible outliers in the car pricing dataset where the selling price is significantly higher or lower than the market price."

• **Prompt 7:**
"Suggest DAX measures for calculating average sale price, maximum price, minimum price, and total vehicle count for a car sales dashboard."

• **Prompt 8:**
"Provide recommendations to improve the visual appearance and layout of a car pricing dashboard in Power BI."

Appendix D — Screenshot Gallery

This appendix contains additional screenshots and visual references that support the analysis presented in the main chapters of the report. These visuals illustrate the dashboard layout, charts, and key insights derived from the car pricing dataset.

Screenshot 1 — Car Price Analysis Dashboard Overview

This screenshot shows the **main Power BI dashboard** created for the project. The dashboard provides a complete overview of the car pricing dataset and includes key performance indicators such as:

- **Total Brand:** Displays the number of unique car brands available in the dataset.
- **Average Sale Price:** Shows the average selling price of vehicles calculated using the dataset.
- **Model Count:** Displays the number of car models analyzed in the dataset.

Screenshot 2 — Vehicle Production by Car Brand

This visual presents a **bar chart** showing the distribution of vehicles across different car brands. It helps identify which brands appear most frequently in the dataset and which brands have lower representation.

Screenshot 3 — Selling Price vs Odometer

This **scatter chart** illustrates the relationship between the **vehicle odometer reading (distance traveled)** and the **selling price**. The visualization helps identify trends such as whether vehicles with higher mileage tend to have lower selling prices.

Screenshot 4 — Vehicle Market Price vs MMR by Brand

This chart compares **vehicle market price with the Manheim Market Report (MMR) value** across different car brands. It helps evaluate how closely the selling price aligns with the estimated market value.

Screenshot 5 — Number of Vehicles Sold by Brand

This **donut chart** shows the proportion of vehicles sold by each car brand. It helps highlight the most popular brands within the dataset.

Screenshot 6 — Geographic Distribution of Vehicles

The **map visualization** displays the geographic distribution of vehicles across different regions. This helps identify where the vehicles in the dataset are primarily located or sold.